

SMART INDIA HACKATHON 2026



NAWI-ReportPro: OIML R-76 Digital Platform

Problem Statement ID: 26035

Problem Statement Title:

Automated Test Report Generator for Non-Automatic Weighing Instruments (NAWI)

Theme: Smart Automation

PS Category: Software

Team ID: SIH2026-TEAM-NAWI

Team Name: Metrology Innovators



NAWI-ReportPro: Automated Digital Metrology Platform

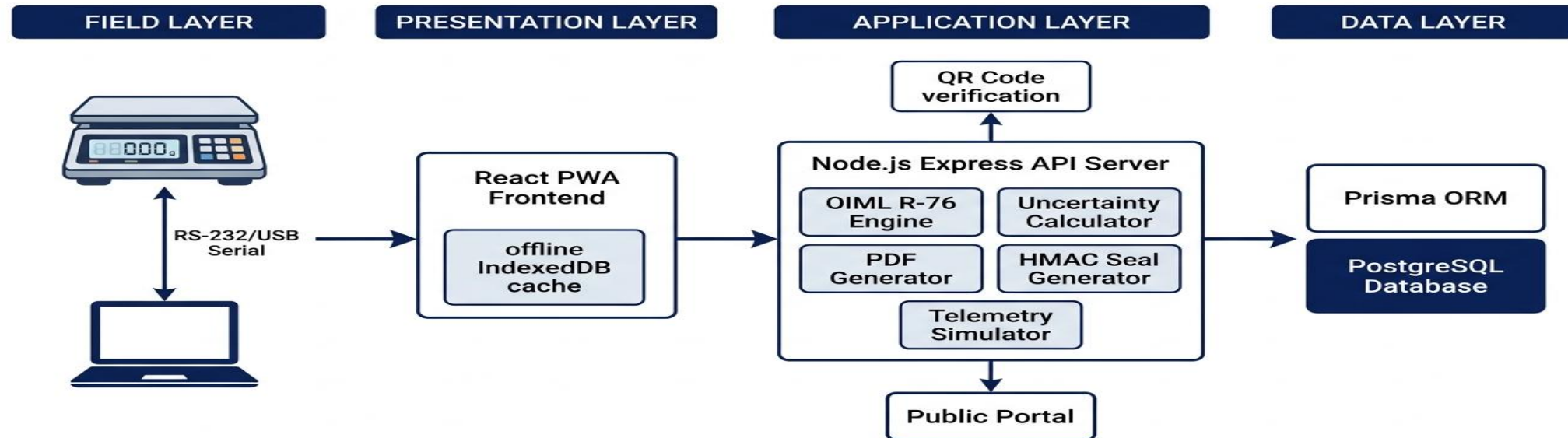
Proposed Solution & Core Innovations

- **Digital Transformation:** Replaces manual paper verification for Non-Automatic Weighing Instruments (scales & truck weighbridges) with an end-to-end digital system.
- **Direct Scale Telemetry:** Connects directly to weighing indicators via browser Web Serial API (RS-232/USB) — live weight capture with zero manual typing errors.
- **OIML R-76 Engine:** Automates Maximum Permissible Error (MPE) calculations across all 4 accuracy classes (Class I to IIII) with multi-interval tare compensation.
- **Tamper-Proof Certificates:** Issues legal verification certificates secured with HMAC-SHA256 digital seals and instant QR code public verification.



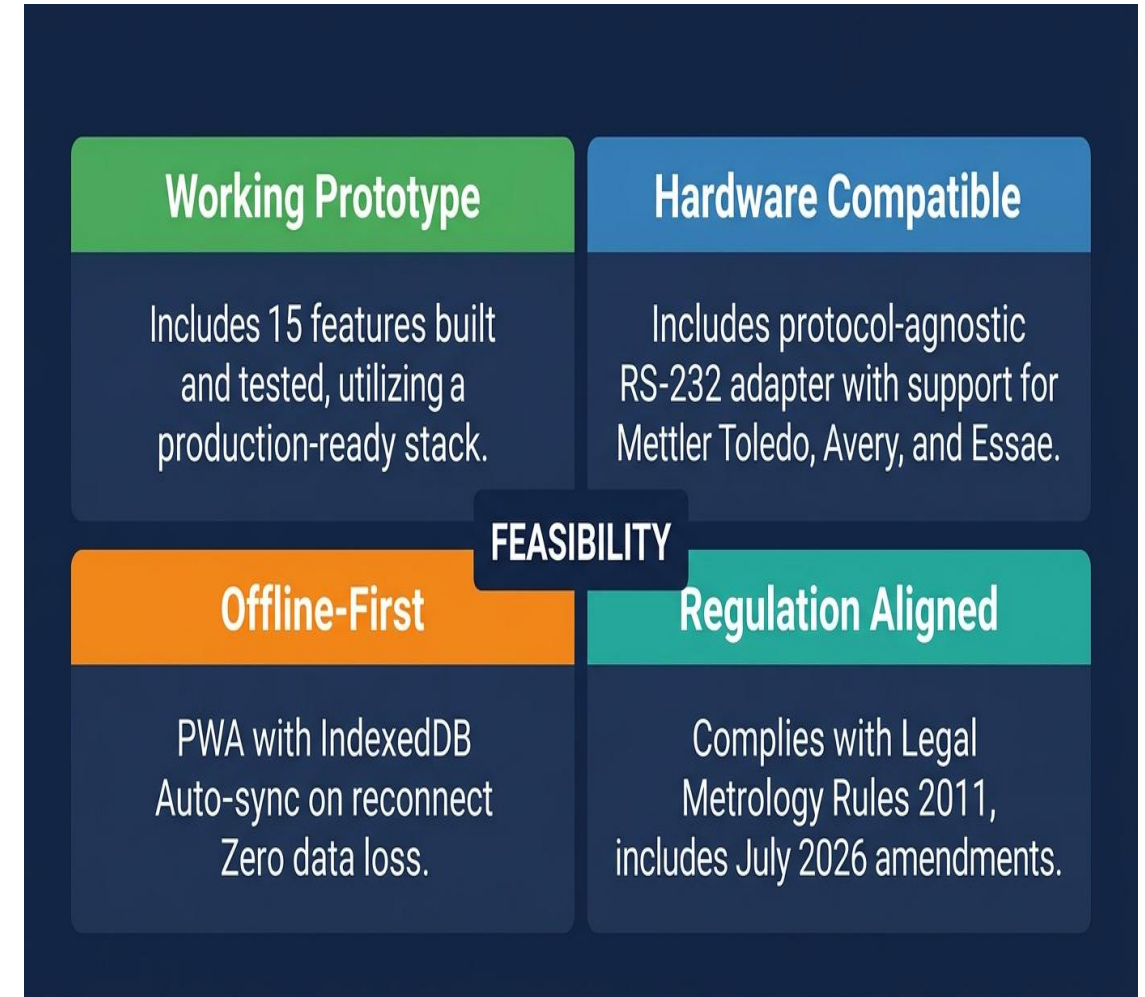
TECHNICAL APPROACH & SYSTEM ARCHITECTURE

- **Hardware Layer:** Browser Web Serial API reads continuous ASCII/HEX telemetry streams from Mettler Toledo (SICS), Avery Weigh-Tronix, and Essae indicators.
- **Application Stack:** React 18 PWA frontend + Node.js Express REST API + PostgreSQL 14 with Prisma ORM + PDFKit high-resolution certificate engine.
- **Metrology & Offline Stack:** OIML R-76 MPE computation engine, ISO GUM uncertainty budget ($k=2$), and IndexedDB queue with idempotent background sync for remote mandis.



Key Feasibility Pillars & Risk Strategies

- **Working Prototype Built:** All 15 core metrological features fully implemented and validated with 100% automated test pass rate (129/129 test cases passing).
- **Hardware Agnostic:** Protocol-agnostic serial parser interfaces with diverse vendor hardware; virtual simulator enables pre-deployment field training.
- **Remote Mandi Resilience:** Offline-first PWA with IndexedDB local caching allows full inspections in zero-connectivity rural yards with auto-sync upon reconnection.
- **Legal Metrology Compliance:** Conforms to OIML R-76 (Edition 2006/E) and the Legal Metrology (General) Rules 2011, including July 2026 amendments for high-capacity scales.



QUANTIFIED IMPACT & NATIONAL BENEFITS

- **10x Faster Field Verification:** Automated calculations reduce inspection duration from 2 hours to under 15 minutes per instrument, eliminating backlogs.
- **Protecting 1.80 Crore Farmers:** Guarantees accurate weighing across 1,656 e-NAM agricultural mandis nationwide (DoCA/e-NAM official data, June 2026).
- **100% Tamper-Proof Legal Seals:** HMAC-SHA256 digital seals with instant public QR validation completely eliminate forged paper calibration certificates.
- **Paperless E-Governance:** Replaces multi-page carbon inspection forms with centralized, cryptographically auditable digital records across all State Directorates.

GOVERNMENT DIGITAL VERIFICATION PLATFORM: TRANSFORMING WEIGHING INSTRUMENTS



**10x
FASTER**
Verification
Speed



**1,656
MANDIS**
Agricultural
Markets Covered



**1.8
CRORE**
Farmers
Protected



100%
TAMPER-PROOF
Digital Seal
Security

MANUAL: 2 hours per instrument

DIGITAL: Under 15 minutes

1. International Metrology Standards

- **OIML R 76-1:2006:** Non-automatic weighing instruments — Metrological & technical requirements; Table 3 MPE tolerance limits, Section 3.5.
- **OIML R 76-2:2007:** Non-automatic weighing instruments — Standardized test report format and repeatable testing procedures.
- **ISO/IEC Guide 98-3:2008 (GUM):** Guide to the Expression of Uncertainty in Measurement — Expanded uncertainty budget calculation ($k=2$).
- **EURAMET cg-18 v4.0 (2015):** Guidelines on Calibration of NAWI — Repeatability, eccentricity & temperature formulas.

2. Indian Legal Framework & Government Portals

- **Legal Metrology Act, 2009 (Act 1 of 2010):** Statutory mandates for verification & calibration of commercial weighing instruments.
- **Legal Metrology (General) Rules, 2011:** Seventh Schedule standards, including July 2026 amendments rationalizing verification norms for high-capacity scales (≥ 1 tonne).
- **BIS IS 9281 (Parts 1-4):** Bureau of Indian Standards specifications for electronic weighing systems and load cells.
- **DoCA eMaap Portal (emaap.gov.in):** Department of Consumer Affairs online enforcement and inspection tracking system.
- **e-NAM Platform (enam.gov.in):** National Agriculture Market official data: 1,656 mandis, 1.80 Cr farmers (June 2026).
- **SIH Winner Benchmark:** <https://github.com/Aadiii00/SIH-Winners-PPT-and-Sources>.